

How good is Rust at compiling C?

Leon McQueen
Hampton School, L6DCW

1. Introduction

The compiler made for this project was built in Rust, a relatively new programming language, with an initial stable release being published “only” eleven years ago, in May of 2015, and a latest stable release of Rust 1.97.1 as of July 21, 2026 [1], [2].

Rust describes itself as “blazingly fast and memory-efficient”, due to its intermediate compilation to the LLVM IR [3] and its unique feature of lifetimes, ensuring memory safety and efficiency throughout the runtime of a program.

2. bits and pieces

Having built an operating system in Rust previously, I was interested to explore its capabilities further. I had recently heard of Rust being integrated into the Linux kernel, and through several YouTube videos had discovered more and more of the capabilities of the language, which piqued my interest of the other capabilities of the language, outside what I had learnt while building the operating system. Additionally, I had been increasingly interested in compilers and machine code (through a brief attempt at building a bootloader in assembly for the operating system)

Bibliography

- [1] The Rust Core Team, “Announcing Rust 1.0.” [Online]. Available: <https://blog.rust-lang.org/2015/05/15/Rust-1.0/>
- [2] The Rust Release Team, “Announcing Rust 1.97.1.” [Online]. Available: <https://blog.rust-lang.org/2026/07/16/Rust-1.97.1/>
- [3] “Overview of the compiler.” [Online]. Available: <https://rustc-dev-guide.rust-lang.org/overview.html>